CASE Study 1: How does a bike-share navigate speedy success?

Welcome to the Cyclistic bike-share analysis case study! In this case study, you work for a fictional company, Cyclistic, along with some key team members. In order to answer the business questions, fo low the steps of the data analysis process: Ask, Prepare, Process, Analyze, Share, and Act. Along the way, the Case Study Roadmap tables — including guiding questions and key tasks — wil help you stay on the right path.

Scenario

You are a junior data analyst working on the marketing analyst team at Cyclistic, a bike-share company in Chicago. The director of marketing believes the company's future success depends on maximizing the number of annual memberships. Therefore, your team wants to understand how casual riders and annual members use Cyclistic bikes differently. From these insights, your team wi l design a new marketing strategy to convert casual riders into annual members. But first, Cyclistic executives must approve your recommendations, so they must be backed up with compeling data insights and professional data visualizations.

Characters and teams

- **Cyclistic:** A bike-share program that features more than 5,800 bicycles and 600 docking stations. Cyclistic sets itself apart by also offering reclining bikes, hand tricycles, and cargo bikes, making bike-share more inclusive to people with disabilities and riders who can't use a standard two-wheeled bike. The majority of riders opt for traditional bikes; about 8% of riders use the assistive options. Cyclistic users are more likely to ride for leisure, but about 30% use the bikes to commute to work each day.
- **Lily Moreno:** The director of marketing and your manager. Moreno is responsible for the development of campaigns and initiatives to promote the bike-share program. These may include email, social media, and other channels.

- **Cyclistic marketing analytics team:** A team of data analysts who are responsible for colecting, analyzing, and reporting data that helps guide Cyclistic marketing strategy. You joined this team six months ago and have been busy learning about Cyclistic's mission and business goals—as wel as how you, as a junior data analyst, can help Cyclistic achieve them.
- **Cyclistic executive team:** The notoriously detail-oriented executive team wil decide whether to approve the recommended marketing program.

About the company

In 2016, Cyclistic launched a successful bike-share offering. Since then, the program has grown to a fleet of 5,824 bicycles that are geotracked and locked into a network of 692 stations across Chicago. The bikes can be unlocked from one station and returned to any other station in the system anytime.

Until now, Cyclistic's marketing strategy relied on building general awareness and appealing to broad consumer segments. One approach that helped make these things possible was the f lexibility of its pricing plans: single-ride passes, fu l-day passes, and annual memberships. Customers who purchase single-ride or ful-day passes are referred to as casual riders. Customers who purchase annual memberships are Cyclistic members.

Cyclistic's finance analysts have concluded that annual members are much more profitable than casual riders. Although the pricing flexibility helps Cyclistic attract more customers, Moreno believes that maximizing the number of annual members wil be key to future growth. Rather than creating a marketing campaign that targets al-new customers, Moreno believes there is a solid opportunity to convert casual riders into members. She notes that casual riders are already aware of the Cyclistic program and have chosen Cyclistic for their mobility needs.

Moreno has set a clear goal: Design marketing strategies aimed at converting casual riders into annual members. In order to do that, however, the team needs to better understand how annual members and casual riders differ, why casual riders would buy a membership, and how digital media could affect their marketing tactics. Moreno and her team are interested in analyzing the Cyclistic historical bike trip data to identify trends.

Three questions wil guide the future marketing program:

1. How do annual members and casual riders use Cyclistic bikes differently?

2. Why would casual riders buy Cyclistic annual memberships?

3. How can Cyclistic use digital media to influence casual riders to become members?

Moreno has assigned you the first question to answer: How do annual members and casual riders use Cyclistic bikes differently? You wil produce a report with the folowing deliverables:

1. A clear statement of the business task

2. A description of al data sources used

3. Documentation of any cleaning or manipulation of data

4. A summary of your analysis

5. Supporting visualizations and key findings

6. Your top three recommendations based on your analysis Use the folowing Case Study Roadmap as a guide. Note: Completing this case study within a week is a reasonable goal.

ASK - Defining the Business Task

What is the problem you are trying to solve?

Analyze 12 months of Cyclistic trip data to uncover behavioral differences between **annual members** and **casual riders**. then provide actionable recommendations to help the marketing team design a targeted digital campaign that encourages casual riders to convert.

Key Stakeholders:

- **Cyclistic Executive Team** – makes final decisions on marketing strategy and budget.
- **Lily Moreno (Marketing Director)** – builds campaigns based on the insights from this analysis.
- **Analytics Team** – responsible for data preparation, analysis, and reporting (my role).
- **Cyclistic Users** – the target audience whose behavior directly informs the strategy.

How can your insights drive business decisions?

My analysis answers three critical business needs:

1. **Understand When, Where, and How Each Group use Cyclistic**: reveals commuting and recreational behavior.

2. **Discover User Motivations and Pain Points**: explores personalized and behavior-based marketing messaging.
3. **Enable Data-Driven Strategy**: supports leadership in making decisions grounded in evidence rather than assumptions.

PREPARE - Understanding and Validating the Data

Data Used

In this casestudy, we will analyze the provided data: *The previous 12 months of Cyclistic trip data (March 2025 → February 2026).*

We should understand the concept: **ROCCC** (Reliable – Original – Comprehensive – Current – Cited), our data is qualified for this concept, conducted by Motivate – the unit operates Bike-share system in real world. Original from Bike-share Chicago. Data includes efficient information: timestamp, locations, user type. No sensitive data stored.

This is the nearest data (2/2026), cited by Motivate International Inc.

How are you addressing licensing, privacy, security, and accessibility?

Data source Description: For this analysis, I used the 12 months of historical trip data from the Divvy bike-share systems in Chicago. The data is provided by Lyft Cyclistic and Scooters under a public data agreement.

The datasets consists of 12 csv files, including ride_id, timestamps, stations, bike type, and user category (member vs. casual), no sensitive information is included in the data, ensuring compliance with privacy and safety requirements.

How does it help you answer your question?

This data have user category, trip duration to track user behaviors/demands by each hour/day/week. Bike type to track the behaviors choosing bike type. Start station and end station to track commuting habits. These columns give me efficient capabilities to analyze the usage behavior of the two customers groups.

Are there any problems with the data?

Columns station has missing values.

Ended_at and started_at are too close to each other.

Cannot analyze demography

PROCESS - Data Cleaning and Preparation (BigQuery Workflow)

Because the combined dataset exceeded spreadsheet limits (>5.4 million rows), I used **Google BigQuery** for scalable and reproducible data processing.

*I downloaded the most recent 12 months of Divvy (Cyclistic) trip data, covering **March 2025 to February 2026**.*

Step Taken:

1. Load all 12 CSVs into BigQuery

Since each CSV exceeded spreadsheet limits, I uploaded them directly to **BigQuery** using the CLI for fast upload (Google Cloud CLI):

```
bq load --source_format=CSV --autodetect cyclistic.trips_202503 "path/to/202503-divvy-tripdata.csv"

C:\Users\LENOVO>bq load --source_format=CSV --autodetect cyclistics.tripdata925 "D:\AI Projects\Data analysis\Case study 1\202509-divvy-tripdata.csv"

Upload complete.

Waiting on bqjob_r18fdd18bc2c4201b_0000019cf1117255_1 ... (9s) Current status: DONE
```

Created a dataset called **cyclistics**.

Each month stored in a dedicated table (e.g., tripdata_202503, tripdata_202508) to maintain traceability.

2. Combine all months into a master table

Result: **5,463,875 rows**.

Schema | Details | Preview | Table Explorer Preview | Insights | Lineage | Data Profile | Data Quality

Row	ride_id	rideable_type	started_at	ended_at	start_station_name	start_station_id	end_s
1	1DF6954C36CCE7CF	electric_bike	2025-06-19 23:52:28.225000 UTC	2025-06-19 23:57:49.827000 UTC	Hegewisch Metra Station	CHI00876	null
2	FB73B600288BB0BB	classic_bike	2025-05-19 00:05:57.886000 UTC	2025-05-19 00:12:15.645000 UTC	Hegewisch Metra Station	20215	Comm
3	698E5E825EF3E141	electric_bike	2025-11-27 16:06:37.488000 UTC	2025-11-27 16:09:34.617000 UTC	Hegewisch Metra Station	CHI00876	Public
4	3B86AA55EAFBC102	electric_bike	2025-09-09 19:22:54.933000 UTC	2025-09-09 19:28:03.443000 UTC	null	null	null
5	A041A1981C019692	electric_bike	2025-03-29 11:08:05.064000 UTC	2025-03-29 11:08:53.574000 UTC	null	null	Public
6	049C8DC2353E073A	electric_bike	2025-09-26 19:29:08.916000 UTC	2025-09-26 19:33:21.933000 UTC	null	null	Comm
7	F5AC7A3A992FA314	electric_bike	2025-09-24 23:58:49.034000 UTC	2025-09-25 00:01:38.661000 UTC	Avenue O & 134th St	CHI00843	null
8	7A2E09A4684341BA	electric_bike	2025-09-05 13:53:54.204000 UTC	2025-09-05 14:10:57.339000 UTC	Avenue O & 134th St	CHI00843	null
9	586E72891E2F1EF5	electric_bike	2025-04-22 06:44:16.646000 UTC	2025-04-22 06:45:55.089000 UTC	Avenue O & 134th St	20214	Public
10	106FD76E4F4774E2	classic_bike	2025-08-13 18:09:20.756000 UTC	2025-08-13 18:54:16.786000 UTC	Avenue O & 134th St	CHI00843	Comm
11	C1F3C8BE8C516167	electric_bike	2025-08-13 14:33:48.292000 UTC	2025-08-13 14:51:15.854000 UTC	Avenue O & 134th St	CHI00843	Avenu
12	85A37202F5684D90	classic_bike	2025-08-17 09:02:51.607000 UTC	2025-08-17 09:49:16.408000 UTC	Avenue O & 134th St	CHI00843	Ewing
13	0EC3DA795E6BBBDC	classic_bike	2025-03-27 14:40:02.387000 UTC	2025-03-27 15:02:37.901000 UTC	Baltimore Ave & 133rd St	24407	Baltim
14	61D5761BA4D4C543	classic_bike	2025-06-12 16:59:01.633000 UTC	2025-06-12 17:02:27.812000 UTC	Baltimore Ave & 133rd St	CHI01947	Comm
15	4711750D3B80F963	electric_bike	2025-08-26 23:11:40.548000 UTC	2025-08-27 00:47:52.883000 UTC	Indiana Ave & 133rd St	CHI01933	null
16	D5184CBF80C1DFB5	classic_bike	2025-03-08 14:53:47.516000 UTC	2025-03-09 16:53:42.101000 UTC	Altgeld Gardens	20212	null
17	CE0AA09BE11B2BE6	electric_bike	2025-08-22 04:00:22.307000 UTC	2025-08-22 04:20:43.100000 UTC	Altgeld Gardens	CHI00849	Altgel
18	21BB957C44669F10	electric_bike	2025-07-13 17:00:56.998000 UTC	2025-07-13 17:04:32.865000 UTC	Commercial Ave & 130th St	CHI00828	null
19	6708C7CF82A440BB	electric_bike	2025-09-05 10:26:22.195000 UTC	2025-09-05 10:51:51.455000 UTC	Commercial Ave & 130th St	CHI00828	Comm
20	E29366170A1B0607	electric_bike	2025-09-05 15:59:37.688000 UTC	2025-09-05 16:03:45.258000 UTC	Commercial Ave & 130th St	CHI00828	Hegew
21	496865500659477E	electric_bike	2025-07-03 00:33:14.199000 UTC	2025-07-03 00:33:22.689000 UTC	null	null	null
22	2A4FA48AF77200B9	electric_bike	2025-05-11 17:33:50.363000 UTC	2025-05-11 18:32:08.356000 UTC	Parnell Ave & 127th St	24411	null

Results per page: 50 1 – 50 of 5463875

All 12 files have the same total of columns with consistent schema.

3. Data Integrity Checks

Now, I conducted multiple checks to ensure data integrity.

I created a new column name: ride_length to calculate ended_at – started_at to check if started_at < ended_at.

Check for invalid timestamps

```
SELECT ride_id

FROM cyclistics.tripdata_2025_2026_clean

WHERE started_at IS NULL OR ended_at IS NULL;
```

Result: None found.

Ensure no negative ride durations

```
SELECT ride_id

FROM cyclistics.tripdata_2025_2026_clean

WHERE ride_length < 0
```

Result: 29 rows

Check null timestamps: none found

Detect rides with negative duration: 29 rows

Check schema consistency: all 12 files fully aligned

Check duplicates: none found

4. Clean the dataset

Due to BigQuery Free Tier restrictions (no DELETE), I recreated a clean table:

```
CREATE OR REPLACE TABLE cyclistics.tripdata_2025_2026_clean AS
SELECT *
FROM cyclistics.tripdata_2025_2026_clean
WHERE ride_length >= 0;
```

No encoding error, format error. No duplicated id.

After cleaning, I verified that all ride_length values were positive, weekday values were between 1 and 7, timestamps were valid. The cleaned files were saved into a table "cyclistics.tripdata_2025_2026_clean" for the analysis phase. This process ensures that the dataset is accurate, consistent, and ready for the analysis step.

ANALYZE - Key Insights

After cleaning and combining all 12 months of Cyclistic trip data, I performed analysis to understand how annual members and casual riders use the service differently.

I conducted multiple SQL analyses in BigQuery to uncover usage patterns.

A few unexpected findings emerged:

- A number of rides had negative durations, likely due to device time sync issues.
- Using a rideable-type distribution query:

```
SELECT member_casual,
    rideable_type,
```

```
    COUNT(*) AS total_rides
FROM `cyclistics.tripdata_2025_2026_clean`
GROUP BY member_casual, rideable_type
ORDER BY member_casual, total_rides DESC;
```

Row	member_casual	rideable_type	total_rides
1	casual	classic_bike	670247
2	casual	electric_bike	1318507
3	member	electric_bike	2232794
4	member	classic_bike	1242298

I discovered that members used electric bikes far more frequently than expected. Members recorded over 2.23 million electric-bike trips—significantly higher than the 1.32 million trips by casual riders. This suggests that members rely on electric bikes for faster daily commuting.

- By aggregating ride counts by day of week:

```
SELECT day_of_week,
    member_casual,
    COUNT(*) AS total_rides
FROM `cyclistics.tripdata_2025_2026_clean`
GROUP BY day_of_week, member_casual
ORDER BY day_of_week, member_casual;
```

Row	day_of_week	member_casual	total_rides
1	1	casual	331040
2	1	member	373771
3	2	casual	227565
4	2	member	490715
5	3	casual	223997
6	3	member	552360
7	4	casual	219575
8	4	member	539834
9	5	casual	254973
10	5	member	559456
11	6	casual	319021
12	6	member	517894
13	7	casual	412583
14	7	member	441062

Weekly ride patterns show a weekend spike among casual riders. Casual usage peaks on Saturday with 412,583 rides—nearly double the volume seen during weekdays. In contrast, member usage remains stable Monday through Friday and declines on weekends. This pattern clearly indicates that casual riders use the service primarily for recreational weekend activities (suggesting recreational usage), while members use it for weekday commuting.

- Calculate the average ride length, casual riders take **much longer rides** than members → aligns with recreational behavior. Member usage remains stable across weekdays, suggesting that **members primarily rely on the service for consistent, routine transportation**. **Casual riders demonstrate a weekend-driven pattern where weekend dramatic increase over 35% higher than weekdays, indicates that casual riders use the service mainly for leisure, tourism, and recreational trips.**

Row	member_casual	avg_ride_length_...
1	member	744.8532516549...
2	casual	1360.780823570...

Row	member_casual	day_of_week	avg_duration
1	member	4	712.6587877014...
2	member	5	717.5681286821...
3	member	2	718.7019716128...
4	member	3	720.5823538996...
5	member	6	741.5566351415...
6	member	7	813.8310804376...
7	member	1	825.5641395399...
8	casual	4	1128.269047022...
9	casual	3	1191.161546806...
10	casual	5	1193.765516348...
11	casual	2	1350.932652209...
12	casual	6	1357.673704865...
13	casual	7	1516.502335287...
14	casual	1	1574.098438255...

For further insights, I wil create month columns to explore the usage behavior of the two groups for each month.

First, we will observe the distribution of members for each month

```
SELECT
  month,
  COUNT(ride_id) AS total_rides
FROM `cyclistics.tripdata_with_month`
WHERE member_casual = 'member'
GROUP BY month
ORDER BY month;
```

Row	month	total_rides
1	1	12
2	2	160290
3	3	212268
4	4	262137
5	5	319845
6	6	386795
7	7	440106
8	8	452439
9	9	449294
10	10	422058
11	11	257401
12	12	112447

Member usage increases steadily starting from February, rising month over month until it reaches a peak in August. This pattern suggests that warmer weather strongly influences member ridership: the warmer it gets, the more frequently members use Cyclistic bikes. Ride volume remains high from June through September, indicating a consistent peak season for commuting and daily transportation.

Then, we would observe the distribution of casual for each month

```
SELECT
  month,
  COUNT(ride_id) AS total_rides
FROM `cyclistics.tripdata_with_month`
WHERE member_casual = 'casual'
GROUP BY month
ORDER BY month;
```

Row	month	total_rides
1	1	6
2	2	41178
3	3	85862
4	4	109239
5	5	182770
6	6	292006
7	7	323352
8	8	337878
9	9	265268
10	10	224038
11	11	99082
12	12	28075

Casual ridership shows a strong seasonal trend, with minimal usage in winter and rapidly increasing activity as warmer weather arrives. Ride volume begins to rise significantly in March and continues climbing through the spring, eventually peaking in **August**, which records the highest number of casual rides.

This pattern suggests that casual riders primarily use Cyclistic bikes for **recreational and leisure activities**, which are more common during favorable weather conditions. High ridership from **June to September** reinforces the idea that summer months attract the most casual usage.

Like the member dataset, the January value is artificially low due to **missing or incomplete data**. Therefore, it should not be interpreted as a true reflection of ridership behavior.

As temperatures drop and fall begins, casual ridership declines sharply—especially between September and November—indicating that casual users are significantly more influenced by seasonal conditions than members.

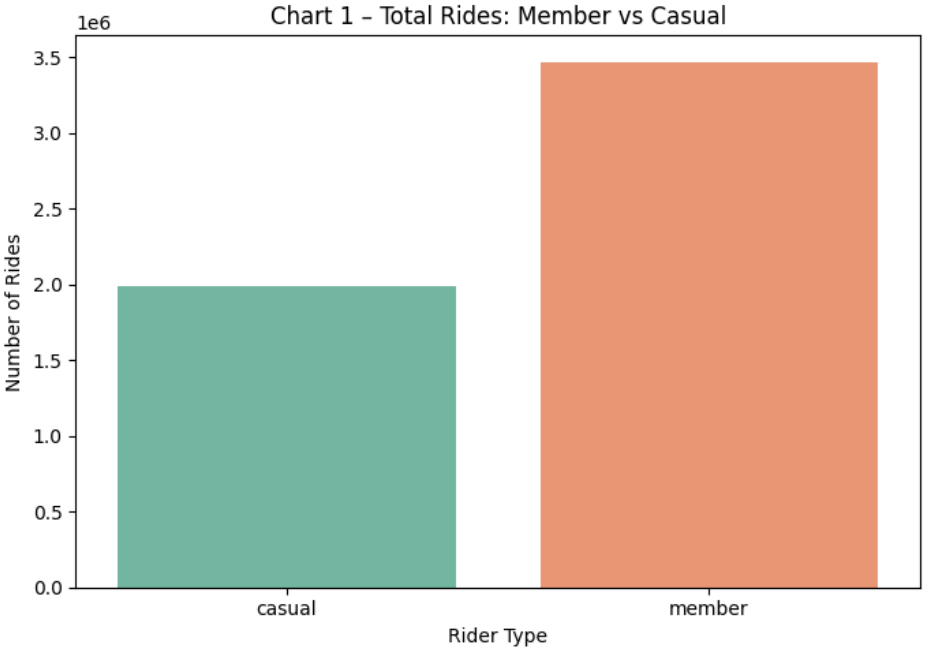
The available data ranges from **December to September**, which means January data is missing. As a result, the unusually low value recorded for January should be interpreted as incomplete rather than an actual drop in ridership.

So, this insight can guide targeted marketing and operational planning. We can promote weekend leisure bundles to casual riders. Adjust bike distribution to match weekend surges.

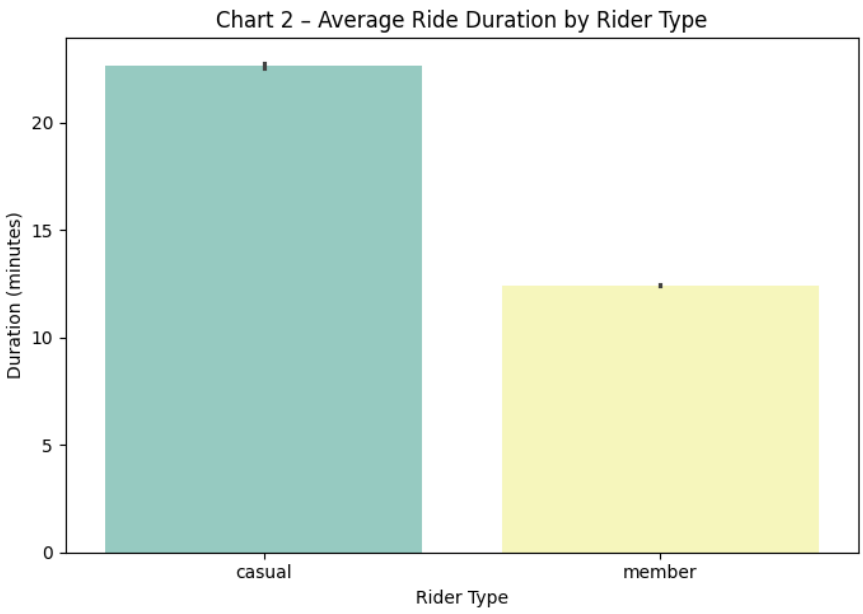
SHARE

After completing the analysis, my next goal was to communicate the insights in a way that the Cyclistic executive team could quickly understand and act upon. Because my audience consists of decision-makers rather than technical analysts, I focused on building clear, business-oriented visual stories that highlight the behavioral differences between annual members and casual riders.

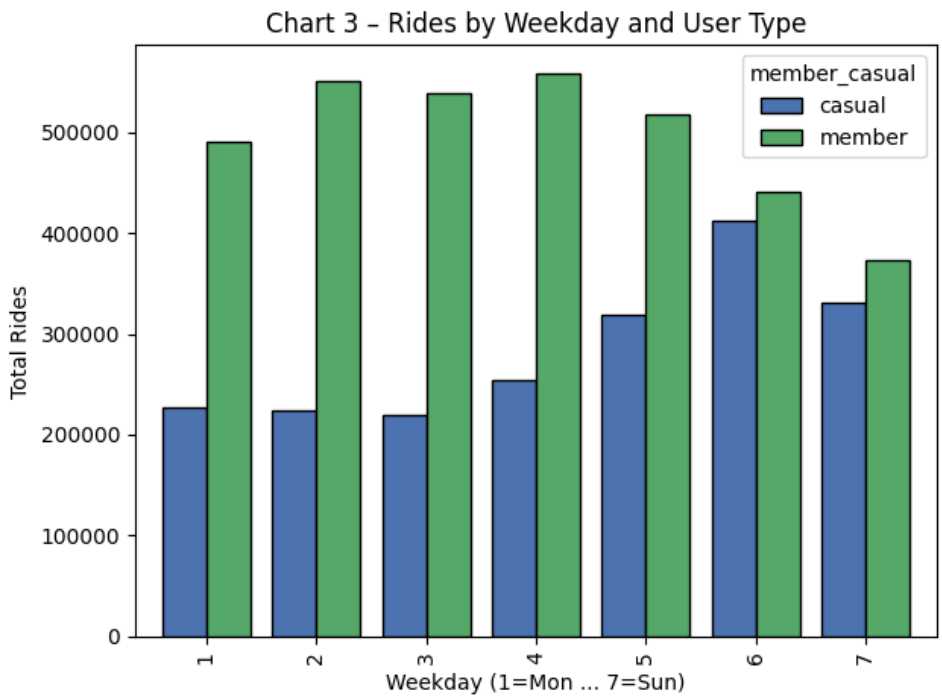
To begin, I spent time sketching different visualization ideas on paper. This helped me decide how to structure the narrative, starting with a broad overview of activity before moving into deeper behavioral comparisons. The first visualization presents the total number of rides across the analysis period, giving the executive team a clear sense of scale and usage patterns.



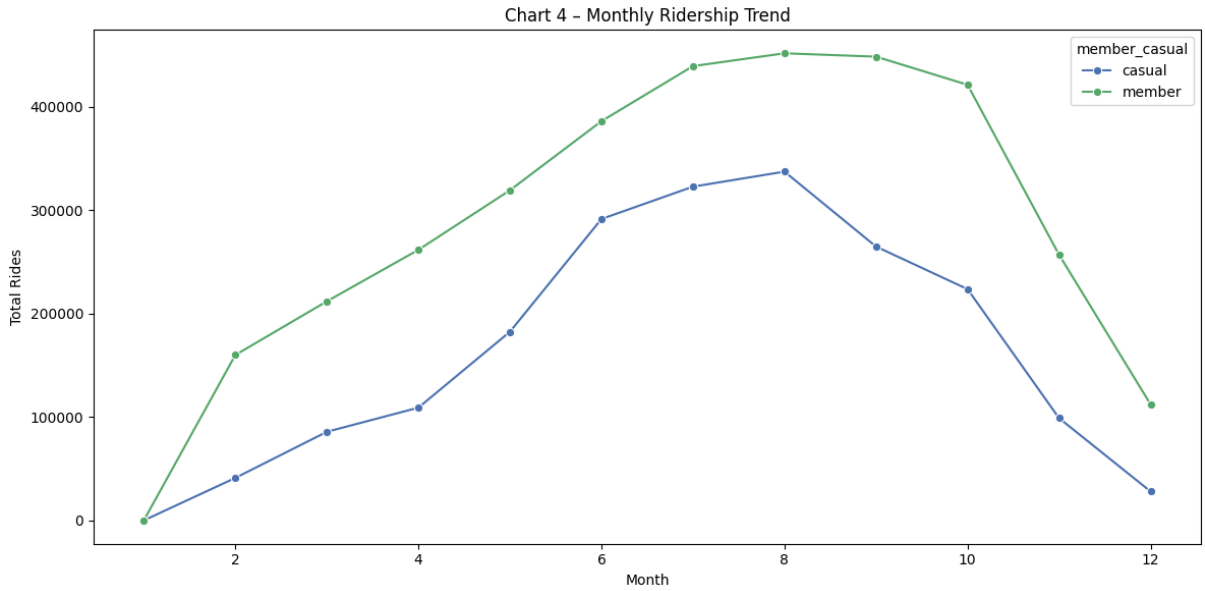
Once the overall trends were established, I focused on ride duration to highlight differences in how each group uses the service. The ride duration distribution shows that casual riders consistently take longer trips, reinforcing their recreational usage pattern, while members are more time-efficient.



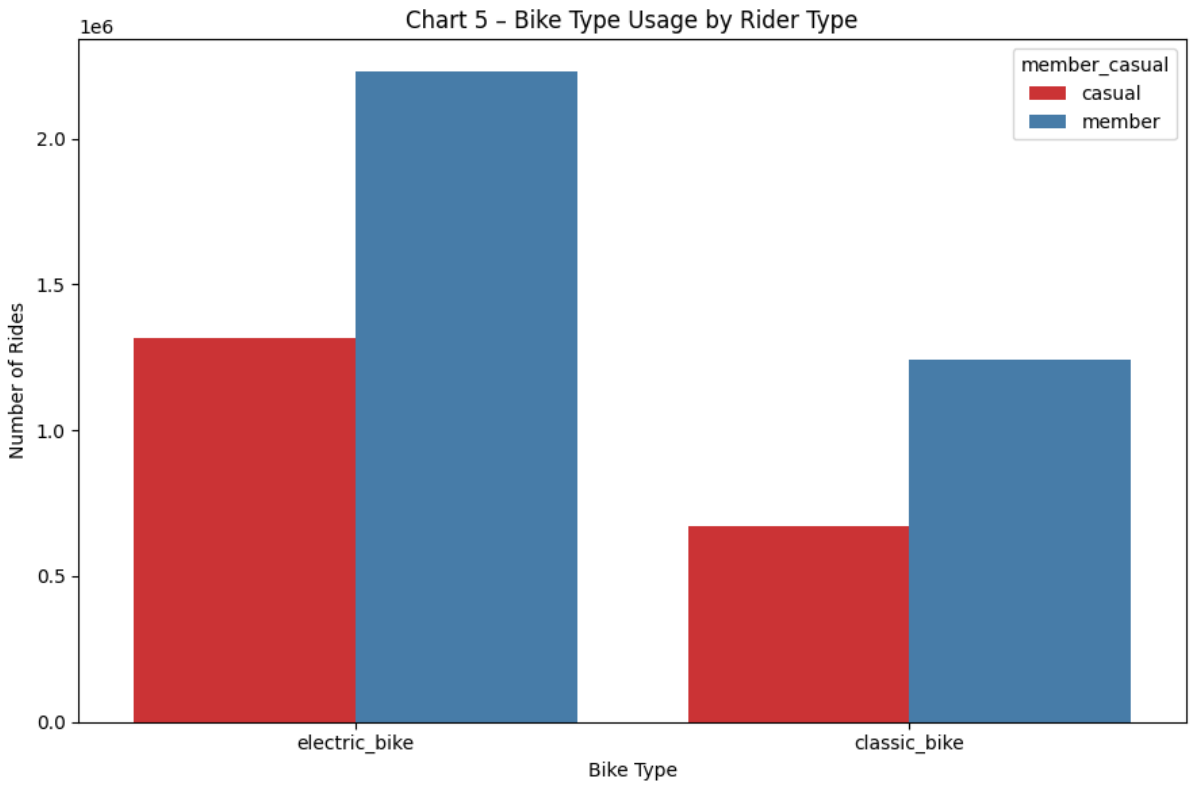
Next, I analyzed weekday versus weekend behavior to identify routine patterns. The resulting visualization clearly illustrates that casual rider activity spikes sharply on weekends, whereas member usage remains stable throughout weekdays, reflecting commuting habits.



To reveal seasonal variations, I examined monthly ridership. This visualization shows the strong effect of warm-weather months on casual riders, who surge in summer, while members maintain more consistent monthly patterns. This further strengthens the distinction between leisure-oriented and commuter behaviors.



Finally, I compared the types of bikes most frequently used by each group. The rideable-type chart reveals that casual riders heavily favor electric and classic bikes, while members show a more balanced distribution. This insight supports operational planning, such as fleet allocation and maintenance prioritization.



Throughout all visualizations, I used clear titles, consistent color themes, and intuitive layout to ensure the presentation is accessible to non-technical stakeholders. Each chart includes a concise headline that communicates the insight at a glance.

Together, the five visuals tell a coherent, data-driven story: casual riders behave like leisure users, riding longer, preferring weekends, and reacting strongly to seasonal changes, while members show steady, shorter, weekday-focused activity aligned with daily commuting. These findings directly support strategic marketing decisions and provide a clear path for targeted conversion efforts.

ACT — Final Recommendations & Portfolio Deliverables

The analysis clearly shows that casual riders and annual members use Cyclistic bikes for fundamentally different purposes. Members demonstrate consistent weekday usage aligned with commuting, whereas casual riders behave like leisure users, favoring weekends, longer rides, and seasonal peaks. These differences highlight specific behavioral levers that Cyclistic can use to convert casual riders into annual members.

So, as a data analyst I would have 3 recommendations for Cyclistic.

We need to focus on membership marketing on weekends since casual riders overwhelmingly ride on weekends and during warm seasons, Cyclistic should launch a weekend membership benefit campaign to attract casual riders. For example, Unlimited rides for spontaneous weekend trips or lower cost ride compared to single passes. This strategy not only remains the loyalty of members but also trigger casual riders' most dominant behavior.

Casual riders consistently take much longer trips, we can use this advantage by promoting messaging like "Your usual 40-minute ride becomes free after the first few minutes with membership". This translates their current spending pattern into a compelling economic reason to convert.

Results:

- Marketing teams can design targeted campaigns based on behaviors, not assumptions.
- Product teams can adjust prompts inside the app (ride-duration nudges, weekend incentives).
- Operations can allocate more electric/classic bikes to high-casual-traffic areas.
- Strategic planning can focus membership growth budgets on the highest-impact months.